Group 6

2026

PROGRAMMING 3D POE PART 1 RESEARCH

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Table of Contents

Table Of Figures

1.1 Introduction

This research investigates existing Android applications that provide functionality like the proposed fitness companion and tracking application. The proposed system will allow users to set personal fitness goals, track nutrition, exercise and weight, monitor long-term progress, follow workout plans and receive reminders and achievements.

Three applications were selected for comparison: *MyFitnessPal*, *FatSecret* and *Calorie Counter by Lose It*. These were chosen because they provide overlapping functionality, particularly food logging, calorie tracking, weight management, exercise tracking and progress monitoring. The research aims to identify effective features and design approaches that can be incorporated into the proposed application while addressing limitations found in existing solutions.

The analysis considers each application's purpose, target users, strengths, weaknesses and likely technical implementation. The findings are then used to establish a prioritised feature strategy for the proposed application.

2.1 Competitive Analysis - App 1 - "MyFitnessPal"

Figure 1: MyFitnessPal App Icon

2.1.1 Overview of the App

MyFitnessPal is a nutrition and fitness tracking application aimed at users who want to manage food intake, exercise, weight and overall fitness. Its Android application provides a food database, exercise tracking, calorie monitoring and progress information. (MyFitnessPal, 2025)

A central feature is its food diary, where users search for foods, select serving sizes and record meals. It also supports barcode scanning and camera-based Meal Scan functionality, although some advanced features require a Premium subscription. Its "Today" dashboard provides an overview of nutrition, exercise, water, habits and progress. (MyFitnessPal, 2025)

2.1.2 Strengths and Weaknesses

- ***Strengths:*** MyFitnessPal's main strength is its extensive nutrition tracking functionality. Users can monitor calories and macronutrients without manually calculating nutritional values. Barcode scanning and Meal Scan also reduce manual data entry and make food logging more convenient. The application provides significant customisation allowing users to configure calorie and macronutrient targets according to their goals. (Hetherton, 2025)
- ***Weaknesses:*** Some useful features, including barcode scanning and Meal Scan, are restricted to Premium users. This may limit accessibility for users who expect essential tracking features to be freely available. Another limitation is the large number of features, which can make the application feel complex for new users. The proposed application could improve on this by providing simpler navigation and prioritising the most important daily actions. (Inkster, 2017)

2.1.3 Technical Implementation

A modern Android implementation could use an MVVM architecture, separating the interface, business logic and data-access layers. View Models could manage screen state, while repositories could communicate with remote APIs and local storage. (Verma and Verma, 2025)

Room could store frequently accessed information such as meals, diary entries and user preferences, while a REST API could manage accounts and centralised nutrition data. Kotlin, Jetpack Compose and Coroutines/Flow could support the interface and reactive state management. Camera and barcode scanning technologies would be required for food recognition and product scanning. (Verma and Verma, 2025)

2.1.4 Interface Architecture

Figure 2 :Progress Overview

This dashboard provides a high-level summary of the user's weekly nutritional performance, displaying 7-day averages for total calories, carbohydrates, fat, and protein. It also tracks weight changes over time, showing starting weight, current weight, and total loss, making it easy to gauge overall progress briefly.

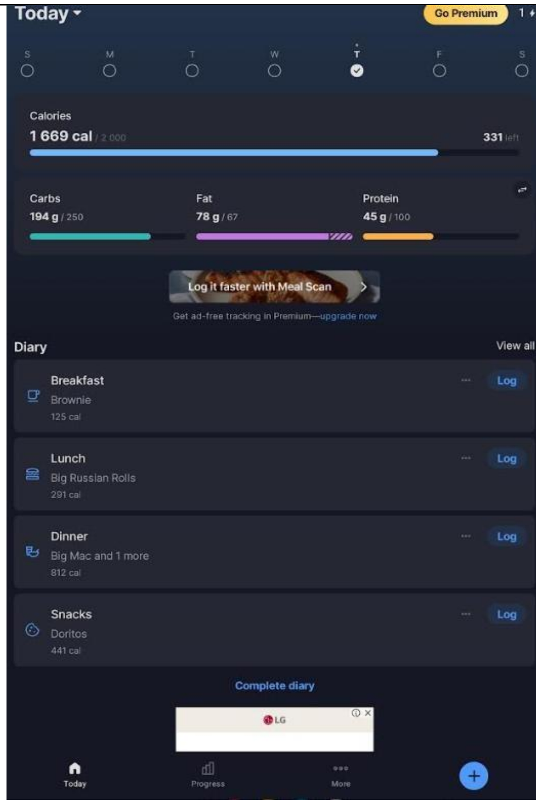


Figure 3: Today's Summary

This screen gives users a clear snapshot of their current daily intake, comparing calories and individual macros (carbs, fat, protein) against personalized daily targets. The visual progress bars quickly show how much of each nutrient has been consumed and what remains, helping users make informed decisions about their next meal.

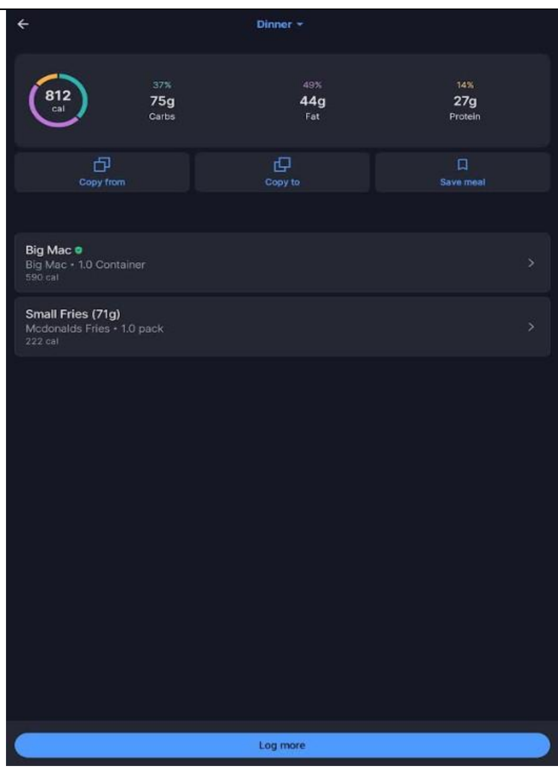
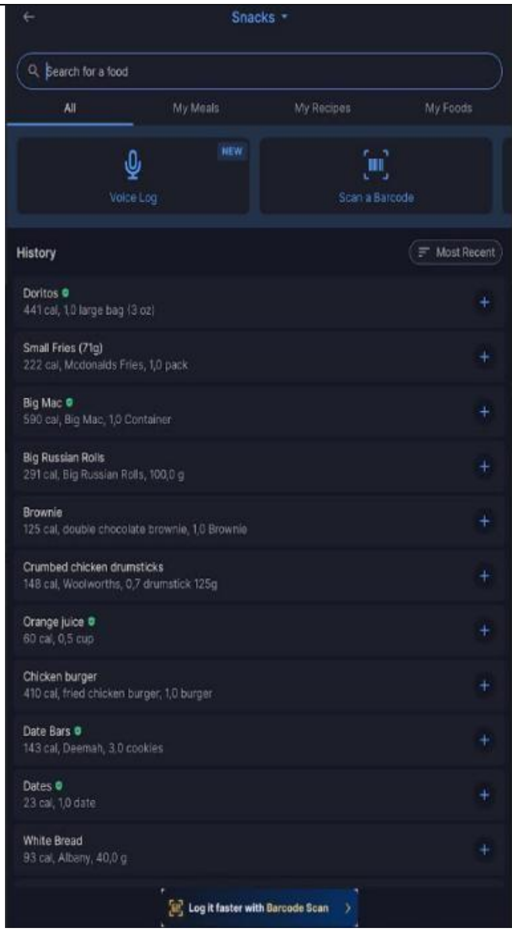


Figure 4: Dinner Details

This detailed meal view breaks down the nutritional composition of a specific dinner, a Big Mac and fries, showing total calories and the percentage contribution from carbs, fat, and protein. It also lists each food item individually with its calorie count, allowing users to understand exactly where their nutrients are coming from within a meal.

	***Figure 5: Snack Logging*** This interface simplifies the process of adding snacks by offering a searchable food database alongside a scrollable history of recently logged items. Users can quickly re-add frequent foods with a single tap or search for new items, while barcode scanning functionality provides an even faster logging option for packaged products.

2.2 Competitive Analysis - App 2 – "FatSecret"

Figure C: FatSecret App Icon

2.2.1 Overview of the App

FatSecret is a nutrition, calorie and weight-tracking application focused on helping users manage food intake, exercise and weight. Its Android application includes a food diary, exercise diary, weight tracking, barcode scanning and progress reports. (fatsecret, 2016)

It is particularly relevant to the proposed application because it combines nutrition, exercise and weight management. It also provides personalised calorie targets and long-term progress information. (fatsecret, 2016)

2.2.2 Strengths and Weaknesses

- ***Strengths:*** Fat Secret's food diary allows users to record meals and monitor calories and nutrients. Food can be added through searching, barcode scanning and image-based recognition. Its barcode scanner is particularly useful because it reduces manual entry and is promoted as a free feature. Another strength is its focus on long-term progress. Weight charts, exercise history, reports, achievements and streaks allow users to monitor consistency and behavioural trends. (CalorieCue Team, 2026)

- ***Weaknesses:*** FatSecret remains primarily focused on nutrition and weight management. The proposed application could differentiate itself by combining these features with structured workout plans and dedicated workout sessions. Like MyFitnessPal, its large feature set may also increase interface complexity, creating an opportunity for a simpler and more focused user experience. (South China Morning Post, 2019)

2.2.3 Technical Implementation

FatSecret could similarly use MVVM with ViewModels managing screens such as the Food Diary, Exercise Diary and Progress sections. Room could provide local storage for meals, foods, workouts and weight records, while a REST API could support accounts, centralised data and synchronisation. (Olga, 2023)

Offline functionality is particularly relevant to the proposed application. Local records could be stored while the device is offline and synchronised with the server once connectivity is restored. Jetpack Compose could provide reusable components for food entries, calorie summaries, progress charts and exercise records. (Olga, 2023)

2.2.4 Interface Architecture

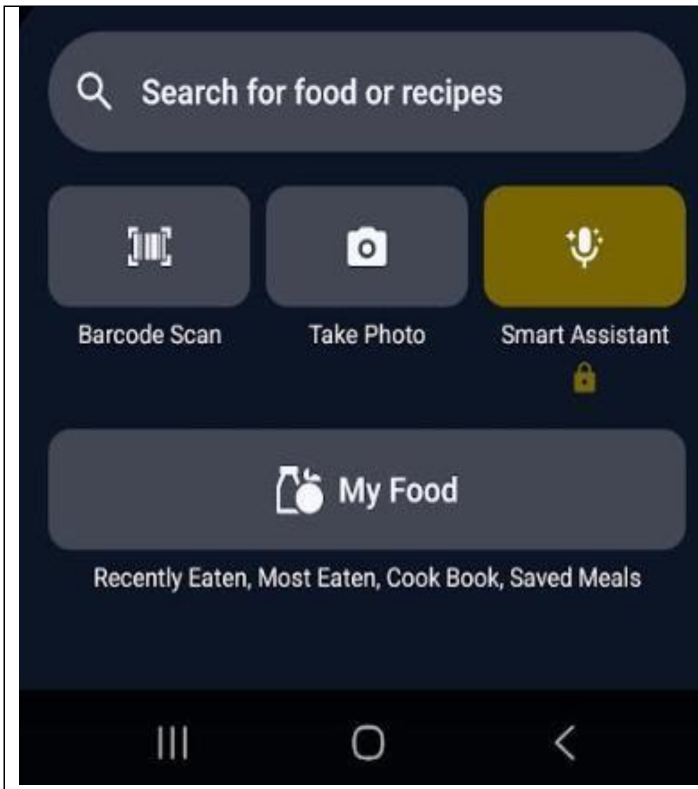


Figure 7: Search Tools

This screen presents users with multiple convenient options for logging food, including manual search, barcode scanning, photo recognition, and a smart assistant. It also organizes saved items into helpful categories like recently eaten, most eaten, cookbook, and saved meals, streamlining the logging process for frequent users.

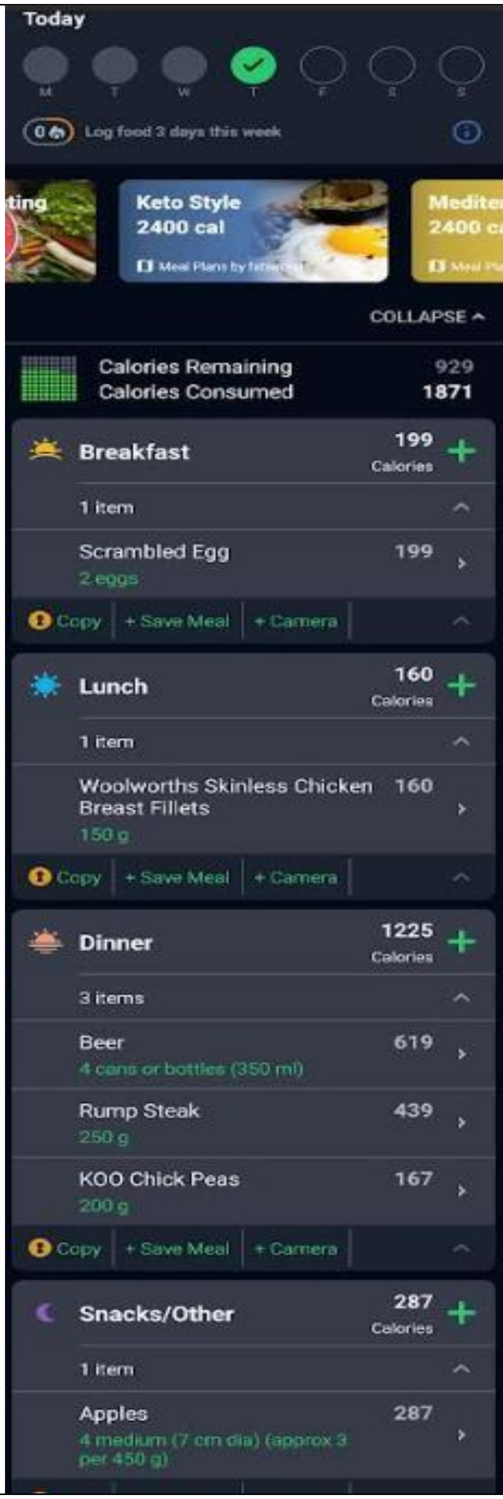


Figure 8: Full Day Meal log diary

This comprehensive diary view displays the user's complete daily food intake, organized by meal categories from breakfast through snacks. Each entry shows portion sizes and provides quick-action buttons for copying, saving meals, or using the camera. The header also promotes meal plan options and tracks weekly logging consistency.

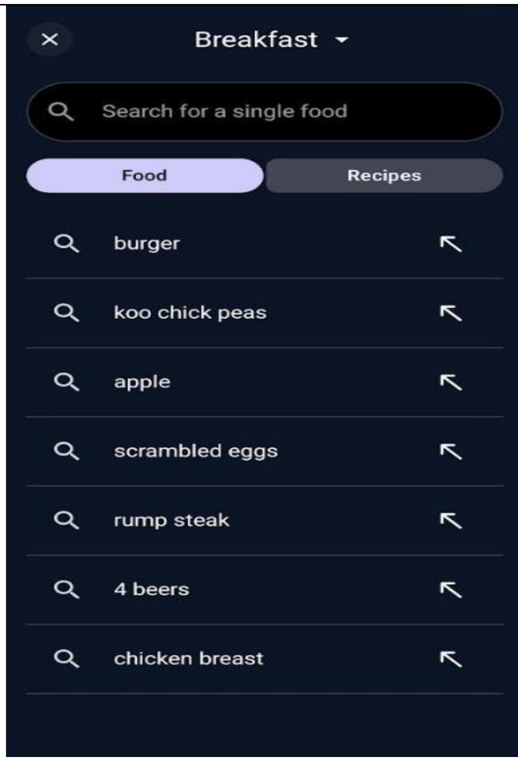


Figure 3: Quick food search

Designed for fast logging, this search interface allows users to find specific foods or browse recipes. It displays recent search history with common items like eggs, steak, and chicken breast, making it easy to quickly select and add frequently eaten breakfast foods without retyping searches each time.

Figure 10: Goals & Calorie Progress

This progress screen visually tracks the user's daily calorie intake against a personalized goal of 2,800 kcal, showing 67% completion. It breaks down calories by meal category: breakfast, lunch, and dinner so that users can see which meals contribute most to their total and adjust accordingly.

Figure 11: Exercise Search Screen

This activity logging interface helps users accurately record their physical activity by offering a searchable database of exercises. It lists various running intensities with corresponding durations and calorie burns, allowing users to select the most accurate option for their workout and ensure precise energy expenditure tracking.

2.3 Competitive Analysis - App 3 - "Calorie Counter by Lose it!"

Figure 12: Calorie Counter by Lose it! App Icon

2.3.1 Overview of the App

Lose It! is a comprehensive calorie counting and weight loss application that has been available since 2008, helping over 40 million users track their dietary habits and achieve their weight management goals (Google Play, 2026). The app operates on a foundational principle of providing users with a personalised daily calorie budget based on their profile details and weight goals, and then enabling them to track their food intake, exercise, and progress within a single interface (Nutrient Metrics Research Team, 2026).

The application's core value proposition lies in its user-friendly approach to weight loss through proven principles of calorie counting and nutritional awareness. Users begin by setting weight goals, after which the app calculates a customised daily calorie budget and weight loss plan based on their unique body composition, activity level, and habits (MyNetDiary, 2026). Beyond basic tracking, Lose It! offers a range of features designed to support sustained behaviour change, including meal planning capabilities, macronutrient tracking (proteins, carbohydrates, fats), and integration with various fitness wearables and devices such as Fitbit, Garmin, and Google Fit (Lose It! Help Center, 2026).

The application distinguishes itself through its strong focus on user engagement and habit formation. It includes gamification elements such as logging streaks, progress milestones, and community challenges that drive retention (AI Calorie Tracker Index, 2026). The app features a best-in-class goal-coaching user interface with weekly check-ins, calorie budget visualisation, and dynamic calorie budget adjustments based on weight trends and exercise logged (MyNetDiary, 2026).

2.3.2 Strengths and Weaknesses

- ***Strengths:*** Lose It! demonstrates notable strengths that contribute to its popularity, including a best-in-class goal-coaching UI that provides clear visual progress tracking and dynamic calorie budgets that adapt daily targets based on weight trends and exercise logging (MyNetDiary, 2026). The app offers deep integration with wearable devices such as Fitbit and Garmin for automatic exercise offset, alongside strong social features including group challenges that drive longer user retention (AI Calorie Tracker Index, 2026). Additionally, Lose It! provides the most affordable Premium upgrade among major calorie-tracking apps at $39.99 per year (Nutrient Metrics, 2026), with a substantial food database and onboarding that is considered the best in the category for building habit mechanics (Nutrient Metrics Research Team, 2026).

- ***Weaknesses:*** The application faces significant limitations, with the AI-powered photo recognition feature being cloud-dependent and demonstrating poor accuracy. An NIH study found that photo-based apps like Lose It! underestimated meal calories by 250 to 345 calories on average and fat by approximately 30 grams per meal (Hengist et al., 2026, presented at NUTRITION 2026). The free tier displays only four nutrients (calories, fat, carbs, protein), locks essential features like the barcode scanner and voice logging behind Premium, and recent UI changes that removed the quick-add button have drawn user complaints (MyNetDiary, 2026). Database accuracy also remains a concern, with crowdsourced entries contributing to a 12.8% median variance from USDA reference values (Nutrient Metrics, 2026).

2.3.3 Technical Implementation Theory

Based on analysis of the app's features and performance characteristics, Lose It! appears to follow a modern Android development approach using the Model-View-ViewModel (MVVM) architectural pattern, supported by the app's recent rewrite of significant codebase portions to "modernize code" and "make the experience faster" (Lose It! Blog, 2026). The application leverages cloud-based backend architecture, evidenced by its AI photo recognition requiring server-side processing for food identification (AI Calorie Tracker Index, 2026).

The app utilizes ML Kit from Google for on-device machine learning capabilities, enabling features like the nutrition label scanner that can recognize information in under one second through the camera view (Google Developers, 2026). The machine learning team employs compressed, quantized TensorFlow Lite models hosted in Firebase, allowing seamless model updates without requiring app updates and enabling A/B testing of model versions (Google Developers, 2026).

Data persistence likely relies on a combination of local storage (RoomDB or SQLite) for caching and offline access, alongside cloud storage for synchronization and user data management (Nutrient Metrics Research Team, 2026). The development stack probably utilizes Firebase for authentication, cloud messaging, and analytics, alongside a custom RESTful API hosted on cloud platforms for managing core tracking functionality (Lose It! Help Center, 2026).

2.3.4 Interface Architecture

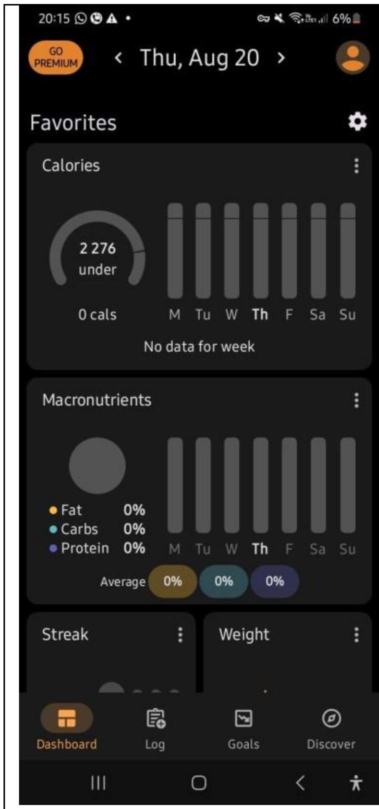


Figure 13: Home/Dashboard Screen (Streak and Summary View)

This screen presents the user's consecutive day streak and a circular progress indicator showing calorie consumption against the daily budget. A weekly calendar view indicates completed logging days, while prominent "Switch to Premium" and "Edit Profile" buttons provide quick access to account management and upgrade options.

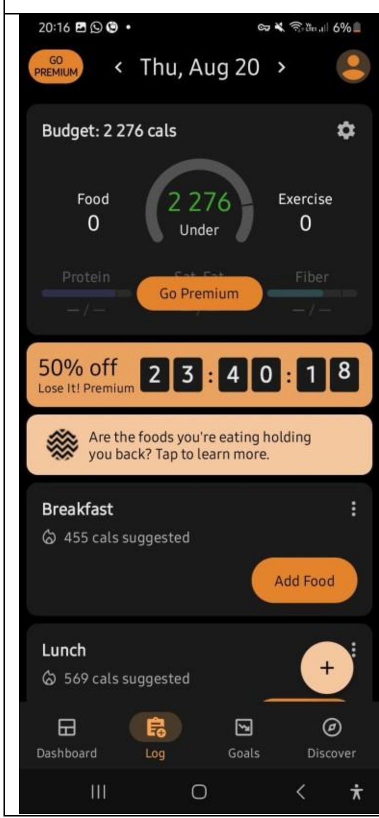


Figure 14: Log Screen (Daily Tracking Interface)

This primary tracking interface displays the current date and remaining calorie budget, with meals arranged sequentially (Breakfast, Lunch, Dinner, Snacks) each featuring an "Add Food" button for easy logging. Cumulative totals for calories and macronutrients appear at the top, while a strategic Premium promotion banner encourages conversions during active use.

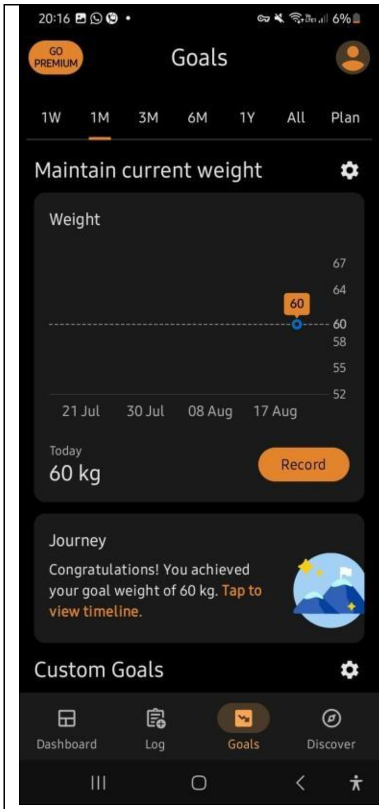


Figure 15: Goals Screen (Progress Tracking and Analytics

This comprehensive progress tracker displays a detailed weight chart with adjustable timeframes (1W to 1Y), shows the current recorded weight with a "Record" button, and celebrates milestones with congratulatory messages upon goal achievement. Additional custom goals can be set, demonstrating the app's flexibility in supporting various health objectives beyond weight management.

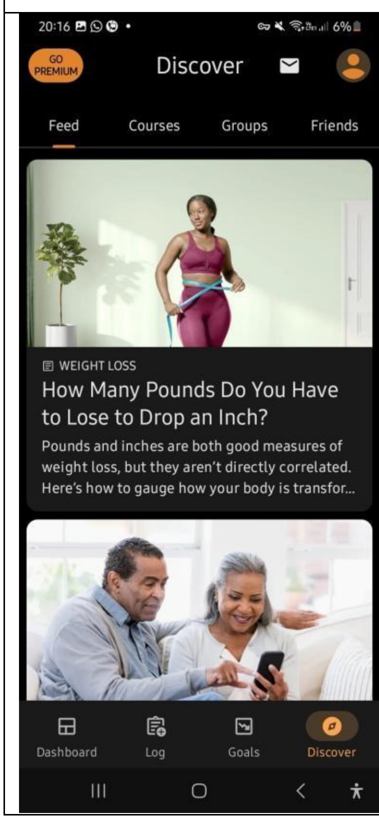


Figure 1C: Discover Screen (Content and Community Hub)

This content hub features a "Go Premium" banner followed by segmented navigation tabs (Feed, Courses, Groups, Friends) that organize community and educational resources. Curated nutrition and weight loss articles appear with preview images and titles, creating an engaging learning and social experience.

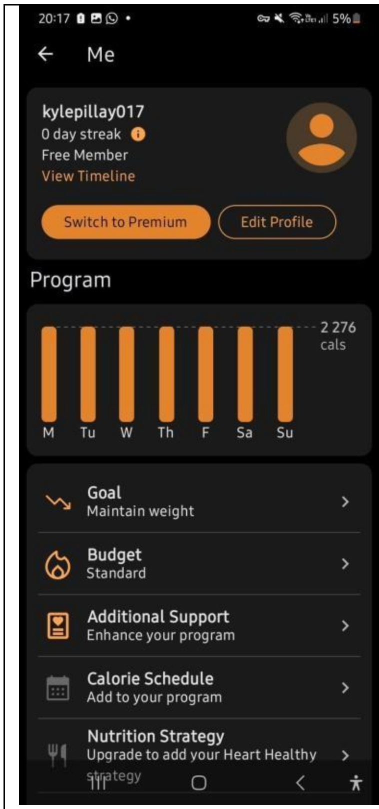


Figure 17: Profile/Edit Profile Screen (User Settings and Preferences)

This screen displays the user's avatar, name, current membership tier (Free Member), and provides access to view their activity timeline. Users can switch to Premium, edit their profile details, and adjust program settings including their weight goal, budget type, and calorie schedule from this centralized settings hub.

3. Comparison Matrix: MyFitnessPal, FatSecret, and Lose It!

Comparison Matrix - Similarities

	MyFitnessPal	FatSecret	Lose It!
Core Functionality	Provides comprehensive calorie tracking, food logging, and weight management tools with a focus on detailed nutritional data and macro/micronutrient analysis.	Delivers essential calorie counting and food logging capabilities with a strong emphasis on community support and user engagement through social features.	Offers goal-oriented calorie tracking and weight loss management with a user-friendly interface and dynamic budget adjustments based on progress.
Database Model	Utilises a crowdsourced food database where users contribute entries, resulting in a vast collection of over 14 million food items with extensive coverage.	Employs a community-driven database model where users add and verify food entries, building a reliable collection through collective contributions.	Leverages a crowdsourced database approach with over 27 million searchable food items, allowing users to log both common and branded products.
Free Tier	Provides an indefinite free version with advertisements, offering basic tracking features and limited nutrient display to encourage Premium upgrades.	Offers a generous indefinite free tier with advertisements, distinguishing itself by providing barcode scanning at no cost to users.	Includes an indefinite free version with advertisements, displaying four basic nutrients while reserving advanced features for Premium subscribers.
Meal Organisation	Structures the day into traditional meal categories including Breakfast, Lunch, Dinner, and Snacks, with the ability to add custom meal names and times.	Organises daily food intake into Breakfast, Lunch, Dinner, and Snacks, allowing users to log meals chronologically throughout the day.	Arranges daily consumption into Breakfast, Lunch, Dinner, and Snacks, with each meal section displaying suggested calorie allocations based on remaining budget.

Comparison Matrix - Differences

	MyFitnessPal	FatSecret	Lose It!
UI Design	Employs a functional, information-dense interface prioritising data visibility with a utilitarian layout that may feel overwhelming to new users but appeals to those seeking detailed nutritional information.	Features a clean, simple interface with a community-focused aesthetic that prioritises user-generated content and social engagement over visual sophistication.	Showcases a modern, polished interface with a spacious, card-based layout designed for simplicity and ease of use, widely regarded as best-in-class for user experience.
Visual Feedback	Provides basic graphs and numerical summaries for progress tracking, with limited motivational elements and standard calorie budget visualisation through simple charts.	Offers straightforward progress indicators and basic visual representations, with less emphasis on dynamic feedback and more focus on community-driven motivation.	Delivers best-in-class visual feedback including circular progress indicators, dynamic calorie budget visualisation, weekly check-ins, and streak tracking that actively motivates user engagement.
Colour Scheme	Utilises a professional blue and white palette that conveys reliability and trustworthiness, with a corporate feel that aligns with its comprehensive, data-driven approach.	Employs a green and neutral colour scheme that suggests health and wellness, creating a calming, organic aesthetic suitable for community-focused interactions.	Features a vibrant, colourful design with warm tones and motivational accents that create an energetic, encouraging atmosphere aligned with goal achievement.
Gamification	Incorporates limited gamification elements including basic streak tracking and progress milestones, with less emphasis on competitive or social motivation features.	Focuses on community-driven gamification through badges, group participation, and social recognition, encouraging user engagement through collective achievement.	Implements strong gamification features including logging streaks, progress milestones, group challenges, and accountability mechanisms that have been shown to drive longer retention rates.

Comparison Matrix - Functional Variations

	MyFitnessPal	FatSecret	Lose It!
Database Entries	Contains over 14 million verified and user-contributed food items, offering extensive coverage for international cuisines and branded products with the largest collection among competitors.	Features a moderate-sized database built through community contributions, sufficient for basic tracking needs but less comprehensive than the other two competitors.	Offers over 27 million searchable food items, the largest database by entry count, though users report accuracy concerns with international foods and crowd-sourced entries.
Barcode Scanner	Locks the barcode scanner feature behind the Premium subscription, a recurring user complaint that forces upgrades for convenient food logging and entry.	Provides the barcode scanner feature within the free tier, distinguishing itself as the most generous free offering among the three competitors.	Restricts barcode scanner functionality to Premium subscribers, requiring payment for this essential convenience feature that competitors may offer for free.
AI Photo Recognition	Offers AI Meal Scan as a Premium feature with cloud-based processing for food identification and portion estimation, though accuracy is not as well documented.	Does not offer AI photo recognition capabilities, relying instead on manual logging and barcode scanning for food entry without machine learning assistance.	Provides AI photo logging as a Premium feature that is cloud-dependent with poor accuracy (68.7% identification rate), slow processing (11.2 seconds), and substantial portion estimation errors (±22%).
Nutrients in Free Tier	Displays only four basic nutrients (calories, fat, carbs, protein) in the free version, with comprehensive macro and micronutrient tracking requiring Premium subscription.	Offers basic nutrient tracking with limited display, focusing on essential information while reserving detailed breakdowns for paid subscribers.	Shows only four nutrients (calories, fat, carbs, protein) in the free version, with advanced tracking requiring the Premium subscription upgrade.

4. Feature Adoption Strategy

Priority	***Feature***	***Justification***
1	Personalised Goals	Use age, height, weight, activity level and fitness goals to calculate personalised calorie and macronutrient targets.
2	Daily Nutrition Dashboard	Display calories consumed/remaining, activity calories and macronutrients in one central location.
3	Food Search C Barcode Scanning	Reduce manual data entry and make food logging faster.
4	Workout Plans C Sessions	Extend beyond calorie tracking by allowing users to follow structured workouts and record sets and repetitions.
5	Progress Analytics	Provide weight, calorie and workout trends through charts and statistics.
C	Offline Mode C Synchronisation	Allow meals, workouts and weight entries to be recorded offline and synchronised when connectivity returns.
7	Notifications C Reminders	Remind users about workouts, weigh-ins and personal goals.
8	Multi-language Support	Support English and at least two South African languages to improve accessibility.
3	Biometric Authentication	Provide fingerprint or facial authentication for convenient and secure access.
10	Settings C Personalisation	Allow users to modify goals, notifications, language, units and preferences.

5. Conclusion

The competitive analysis of ***MyFitnessPal***, ***FatSecret***, and ***Lose It!*** has revealed critical insights into the current state of calorie-tracking and health-monitoring applications. Each application demonstrates unique strengths.

MyFitnessPal excels through its unparalleled database size and comprehensive nutrient tracking. FatSecret distinguishes itself by offering core features like barcode scanning within its free tier and Lose It! leads in user experience design through its exceptional goal-coaching interface, dynamic calorie budgeting, and innovative on-device machine learning capabilities.

However, all three applications share common weaknesses, including database accuracy concerns, paywalled essential features, and limited offline functionality. These findings underscore the importance of balancing feature richness with accessibility, while prioritizing user experience and data reliability.

For our upcoming Planning and Design document, this research establishes a clear technical foundation and feature roadmap. We will adopt the MVVM architectural pattern with a cloud-hosted REST API, implement on-device machine learning using TensorFlow Lite for offline processing, and utilize Firebase for authentication and cloud messaging. The comparison matrix has informed our decision to priorities a dynamic goal-coaching interface, multi-language support for at least two South African languages, and gamification elements to drive user engagement. By incorporating the best identified practices, particularly *Lose Its* approach to user experience and *MyFitnessPal's* database comprehensiveness while addressing the identified shortcomings such as cloud dependency and feature pay walling, we will develop an application that is both technically robust and genuinely user centric.

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